

Noether's Theorem — Derivation

This note derives Noether's theorem in two closely related settings:

- (A) finite-dimensional Lagrangian mechanics (particle systems), and
- (B) classical field theory on spacetime (the standard, general form).

The structure is: assume a continuous one-parameter family of transformations leaves the action invariant (up to a divergence). From the stationarity of the action we build a conserved current and associated conserved charge.

A. Finite-dimensional Lagrangian mechanics

Let the configuration coordinates be $q^i(t)$ with Lagrangian $L(q, \dot{q}, t)$. The action is

$$S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t) dt.$$

Symmetry hypothesis

Suppose there is a one-parameter family of transformations

$$q^i(t) \mapsto q^i(t) + \varepsilon \Delta q^i(q, t) + \mathcal{O}(\varepsilon^2), \quad t \mapsto t + \varepsilon \Delta t(q, t) + \mathcal{O}(\varepsilon^2),$$

under which the Lagrangian changes at most by a total time derivative:

$$L(q + \varepsilon \Delta q, \dot{q} + \varepsilon \Delta \dot{q}, t + \varepsilon \Delta t) = L(q, \dot{q}, t) + \varepsilon \frac{d}{dt} F(q, t) + \mathcal{O}(\varepsilon^2),$$

for some function $F(q, t)$. (When $F = 0$ the Lagrangian is strictly invariant.)

Variation of the action

Compute the first-order change in S under the infinitesimal transformation. Let $\delta q^i = \varepsilon \Delta q^i$ and $\delta t = \varepsilon \Delta t$. The induced variation of the action (keeping fixed integration endpoints in the original parameter t) is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q^i} \delta q^i + \frac{\partial L}{\partial \dot{q}^i} \delta \dot{q}^i + \frac{\partial L}{\partial t} \delta t \right) dt.$$

Integrate the $\delta\dot{q}^i$ term by parts:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}^i} \delta\dot{q}^i dt = \left. \frac{\partial L}{\partial \dot{q}^i} \delta q^i \right|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \delta q^i dt.$$

Collecting terms and using the Euler–Lagrange equations $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0$, the bulk integral vanishes for variations that follow the equations of motion. The remaining boundary contribution plus explicit total-derivative change of the Lagrangian (the F -term) yields

$$\delta S = \left. \frac{\partial L}{\partial \dot{q}^i} \delta q^i \right|_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{d}{dt} (F - L \delta t) dt.$$

Thus, for an on-shell trajectory (satisfying EOM) and when the action is invariant (i.e. $\delta S = 0$ for arbitrary fixed-endpoint symmetry variations), we get

$$\left(\frac{\partial L}{\partial \dot{q}^i} \Delta q^i - F + L \Delta t \right) \Big|_{t_1}^{t_2} = 0.$$

Because the times t_1, t_2 are arbitrary, the quantity in parentheses is constant in time. Define the Noether charge (conserved quantity)

$$Q = \frac{\partial L}{\partial \dot{q}^i} \Delta q^i - F + L \Delta t.$$

Then $dQ/dt = 0$ along solutions of the Euler–Lagrange equations. This is Noether’s theorem in mechanics.

B. Field-theoretic Noether theorem (classical fields)

Let $\phi^a(x)$ be a finite collection of fields on spacetime (coordinates x^μ , $\mu = 0, \dots, d-1$), with Lagrangian density $\mathcal{L}(\phi, \partial_\mu \phi, x)$. The action is

$$S[\phi] = \int_{\Omega} \mathcal{L}(\phi, \partial \phi, x) d^d x.$$

Infinitesimal transformation and Lagrangian variation

Consider an infinitesimal continuous transformation parameterized by ε :

$$x^\mu \mapsto x'^\mu = x^\mu + \varepsilon \Xi^\mu(x) + \mathcal{O}(\varepsilon^2), \quad \phi^a(x) \mapsto \phi'^a(x') = \phi^a(x) + \varepsilon \Delta \phi^a(x) + \mathcal{O}(\varepsilon^2).$$

Equivalently we can describe the field variation at the same spacetime point,

$$\delta \phi^a(x) \equiv \phi'^a(x) - \phi^a(x) = \varepsilon \hat{\Delta} \phi^a(x)$$

where $\hat{\Delta}\phi^a(x) = \Delta\phi^a(x) - \Xi^\mu(x)\partial_\mu\phi^a(x)$ (this subtracts the effect of the coordinate change).

Assume the Lagrangian density changes by at most a total divergence:

$$\delta\mathcal{L} \equiv \mathcal{L}'(x) - \mathcal{L}(x) = \varepsilon \partial_\mu F^\mu(x)$$

for some local $F^\mu(x)$. This is the standard invariance condition: action is invariant (up to boundary terms).

Variation of the action and current construction

Compute the first-order change in the action under the transformation. The variation of the action (for compact domain or decaying fields so boundary terms vanish after including divergence) is

$$\delta S = \int_{\Omega} \left(\frac{\partial\mathcal{L}}{\partial\phi^a} \delta\phi^a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \delta(\partial_\mu\phi^a) + \partial_\mu(\mathcal{L}\Xi^\mu) \right) d^d x,$$

where the $\partial_\mu(\mathcal{L}\Xi^\mu)$ term accounts for the change in the integration measure due to the coordinate shift $x \mapsto x + \varepsilon\Xi$. Expand $\delta(\partial_\mu\phi^a) = \partial_\mu(\delta\phi^a)$ and integrate the derivative term by parts:

$$\int_{\Omega} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \partial_\mu(\delta\phi^a) d^d x = - \int_{\Omega} \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \right) \delta\phi^a d^d x + (\text{boundary}).$$

Collect the bulk terms:

$$\delta S = \int_{\Omega} \left\{ \left(\frac{\partial\mathcal{L}}{\partial\phi^a} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \right) \delta\phi^a + \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \delta\phi^a + \mathcal{L}\Xi^\mu \right] \right\} d^d x.$$

If the fields satisfy the Euler–Lagrange equations,

$$\frac{\partial\mathcal{L}}{\partial\phi^a} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} = 0,$$

then only the total divergence term survives on-shell. Using the invariance assumption $\delta\mathcal{L} = \partial_\mu F^\mu$ we can equate (on-shell)

$$\partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \delta\phi^a + \mathcal{L}\Xi^\mu \right] = \partial_\mu F^\mu.$$

Bring the right-hand side over to the left and define the Noether current

$$J^\mu(x) = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \hat{\Delta}\phi^a + \mathcal{L}\Xi^\mu - F^\mu$$

(notice we used $\hat{\Delta}\phi^a = \delta\phi^a$ the field variation at fixed x). Then the invariance condition implies the local conservation law (on-shell):

$$\partial_\mu J^\mu(x) = 0.$$

This is Noether's theorem: every continuous symmetry (leaving the action invariant up to a divergence) yields a conserved current J^μ . The associated conserved charge is

$$Q = \int_{\Sigma} J^0(x) d^{d-1}x$$

where Σ is a spacelike Cauchy surface; time evolution plus $\partial_\mu J^\mu = 0$ implies $dQ/dt = 0$ (under suitable boundary conditions).

Comments and important special cases

- **Global internal symmetry:** if $\Xi^\mu = 0$ and $\delta\phi^a = \varepsilon \Delta\phi^a$ with $\delta\mathcal{L} = 0$ (strict invariance), then the Noether current reduces to

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \Delta\phi^a.$$

Example: phase rotations of a complex scalar field give electric charge current.

- **Spacetime translations:** take $\Xi^\mu = \varepsilon^\mu$ (constant), $\delta\phi^a = -\varepsilon^\nu \partial_\nu \phi^a$. With $F^\mu = 0$ one obtains the canonical energy-momentum tensor

$$T^\mu_\nu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)} \partial_\nu \phi^a - \delta^\mu_\nu \mathcal{L},$$

and conservation $\partial_\mu T^\mu_\nu = 0$ (Noether currents associated to translations). T^{00} is the energy density; T^{0i} are momentum densities.

- **Gauge (local) symmetries:** if the parameter ε is a function of spacetime, the naive current vanishes on-shell (Noether second theorem) and instead yields identities among the Euler-Lagrange equations (gauge identities). For local gauge symmetries, conserved charges require careful handling (Gauss-law constraints).

Concise proof summary

1. Start with an infinitesimal symmetry transformation of fields and coordinates that changes the Lagrangian by a total divergence.
2. Compute the variation of the action and integrate-by-parts the derivative terms.
3. Use the Euler-Lagrange equations to kill bulk terms; the remaining piece is a spacetime divergence.
4. Define the Noether current as the combination appearing inside the divergence; invariance implies its divergence vanishes on-shell.

5. Integrate J^0 over spatial slice to obtain a conserved charge.

References for further reading: This derivation follows the standard presentations found in many classical mechanics and field theory texts (e.g. Goldstein, Srednicki, Peskin & Schröder, Noether's original paper). For gauge theories and the subtleties of local symmetries see treatments of the Noether second theorem and constrained Hamiltonian systems (Dirac-Bergmann formalism).